Precipitation and Storms

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1 Introduction

Understanding **weather,** the state of the atmosphere (temperature, humidity, precipitation) at a certain place and time, is one of the most difficult parts of learning meteorology. It is the culmination of many other aspects of the subject: Stability, clouds, air pressure, and fronts all play key roles in the formation of rain, snow, sleet, and hail. Weather can also organize itself into storms, such as thunderstorms, extratropical cyclones, and hurricanes. How do all of these processes occur? Read on to find out!

As mentioned above, understanding precipitation and storms is a culmination of all of your other meteorology knowledge, and we highly suggest understanding the previous three handouts: Stability and Cloud Formation, Air Pressure and Winds, and Air Masses and Fronts.

2 Precipitation

Precipitation is any form of water that falls from a cloud towards the ground. However, the water droplets that make up clouds are usually way too small to fall on their own. A normal cloud droplet is only around 10–20 μm in diameter, while raindrops are usually closer to the millimeter scale. So, cloud particles need some way to grow before they can fall as precipitation. There are two main processes that do this: **collision-coalescence** and the **Bergeron process**.

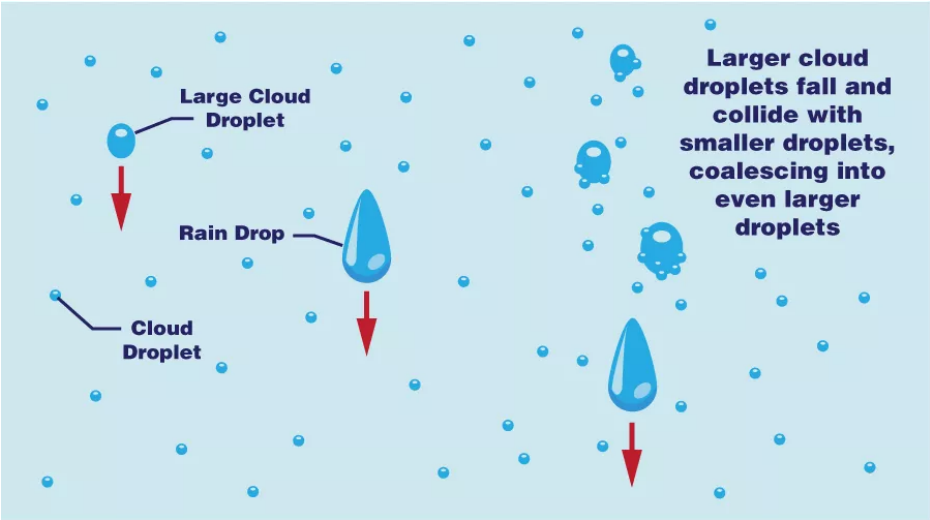
2.1 Formation Processes

2.1.1 Collision-Coalescence

First, we will talk about **collision-coalescence**. This process mainly occurs in **warm clouds**, which are clouds made mostly of liquid water droplets. It is especially important in tropical regions, where clouds can stay above freezing through a large part of their height.

Cloud droplets are not all the same size. Larger droplets have higher **terminal velocities**, meaning that they fall faster than smaller droplets. As these larger droplets fall, they can catch up to and collide with smaller droplets. If the droplets stick together after the collision, they **coalesce** and form one larger drop.

This process is not always efficient. Very small droplets can be carried around a falling drop by the airflow, while droplets that are almost the same size fall at similar speeds and usually do not collide. However, once a few larger **collector drops** form, they can begin collecting smaller droplets much more quickly.



Collision-coalescence in a warm cloud (Source: NOAA NESDIS)

As a collector drop grows, it falls faster and passes through more of the cloud, which gives it even more chances to collide with other droplets. This allows large raindrops to grow pretty quickly once the process gets started.

Updrafts can also help with this process. They keep droplets suspended inside the cloud for a longer amount of time, which gives the droplets more chances to collide and grow. This is one reason why deep clouds with strong vertical motion can produce heavy rainfall.

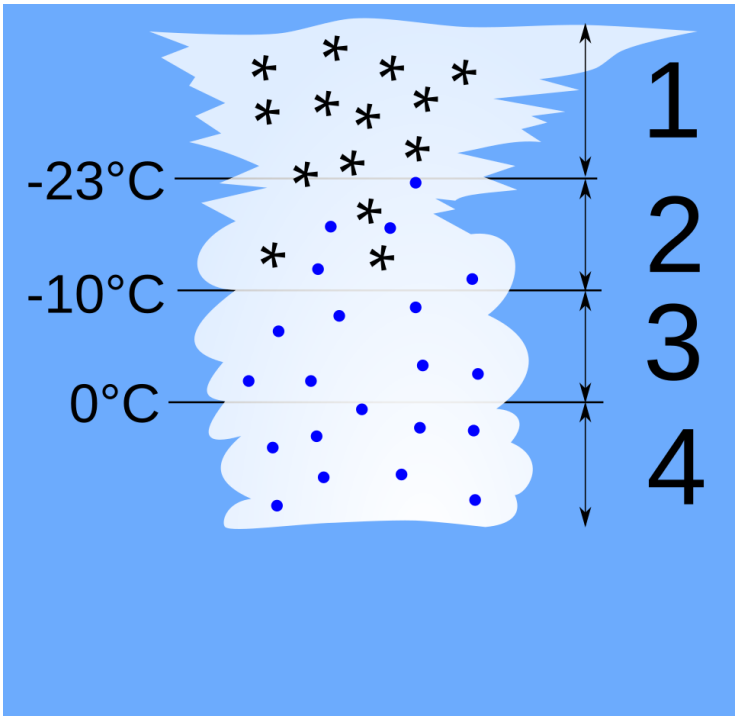
However, raindrops cannot grow forever. As a raindrop gets larger, air resistance begins to flatten it. If it becomes too large, the drop becomes unstable and breaks apart into smaller droplets. Most raindrops therefore do not get much larger than around 5–6 mm.

2.1.2 Bergeron Process

The other major process is the **Bergeron process**, which occurs in colder clouds. One important thing to remember is that water does not always freeze as soon as it reaches 0°C. Liquid water that remains unfrozen below 0°C is known as **supercooled water**. Small cloud droplets can remain supercooled at very low temperatures, although almost all of them freeze by around −40°C.

Because of this, clouds can contain both supercooled water droplets and ice crystals at the same time. These are called **mixed-phase clouds**. This mixture is important because the **saturation vapor pressure over ice is lower than the saturation vapor pressure over liquid water**.

Basically, under the same atmospheric conditions, water vapor can deposit onto ice while nearby liquid droplets lose water through evaporation. Water evaporates from the supercooled droplets and then deposits onto nearby ice crystals. As a result, the ice crystals grow while the liquid droplets shrink. This is the Bergeron process.



The Bergeron process in a mixed-phase cloud (Source: Wikimedia Commons, BoH, CC BY-SA 3.0)

Ice crystals also need something to form on. Particles that help ice form are called **ice nuclei**. Mineral dust and some other particles in the atmosphere can act as ice nuclei. These particles are much less common than cloud condensation nuclei, which means that there are usually many more liquid droplets than ice crystals in a mixed-phase cloud. This actually helps the ice crystals grow because each one has water vapor coming from many nearby droplets.

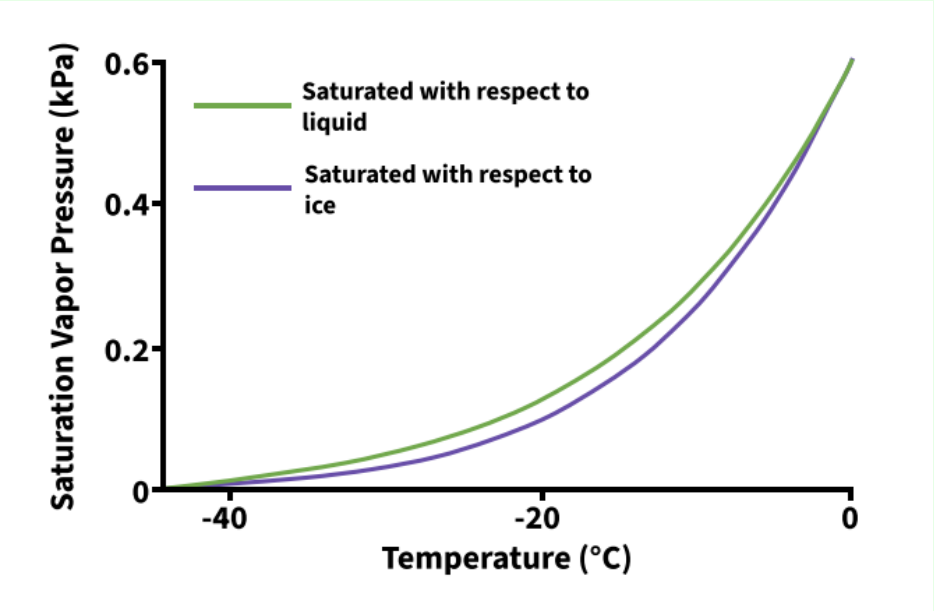
Once an ice crystal forms, it can continue growing in multiple ways:

- **Deposition**: Water vapor changes directly into ice on the surface of the crystal.
- **Aggregation**: Ice crystals collide and stick together, forming larger snowflakes.
- **Riming**: Supercooled water droplets collide with an ice crystal and freeze onto it.

If enough riming occurs, the original ice crystal can become covered in ice and form a rounded particle called **graupel**. Graupel is also important because it can become the starting point for hail inside thunderstorms.

Note: Just because precipitation forms through the Bergeron process does not mean that it will reach the ground as snow. Ice crystals can form high in the cloud and then fall into warmer air. If they melt before reaching the ground, they will fall as rain instead.

USESO Camp Exam 2025, Section I Question 8 The graph below plots saturation vapor pressure as a function of temperature.



Which of the following best explains how the difference between the two curves contributes to precipitation formation in clouds that contain both liquid water droplets and ice crystals?

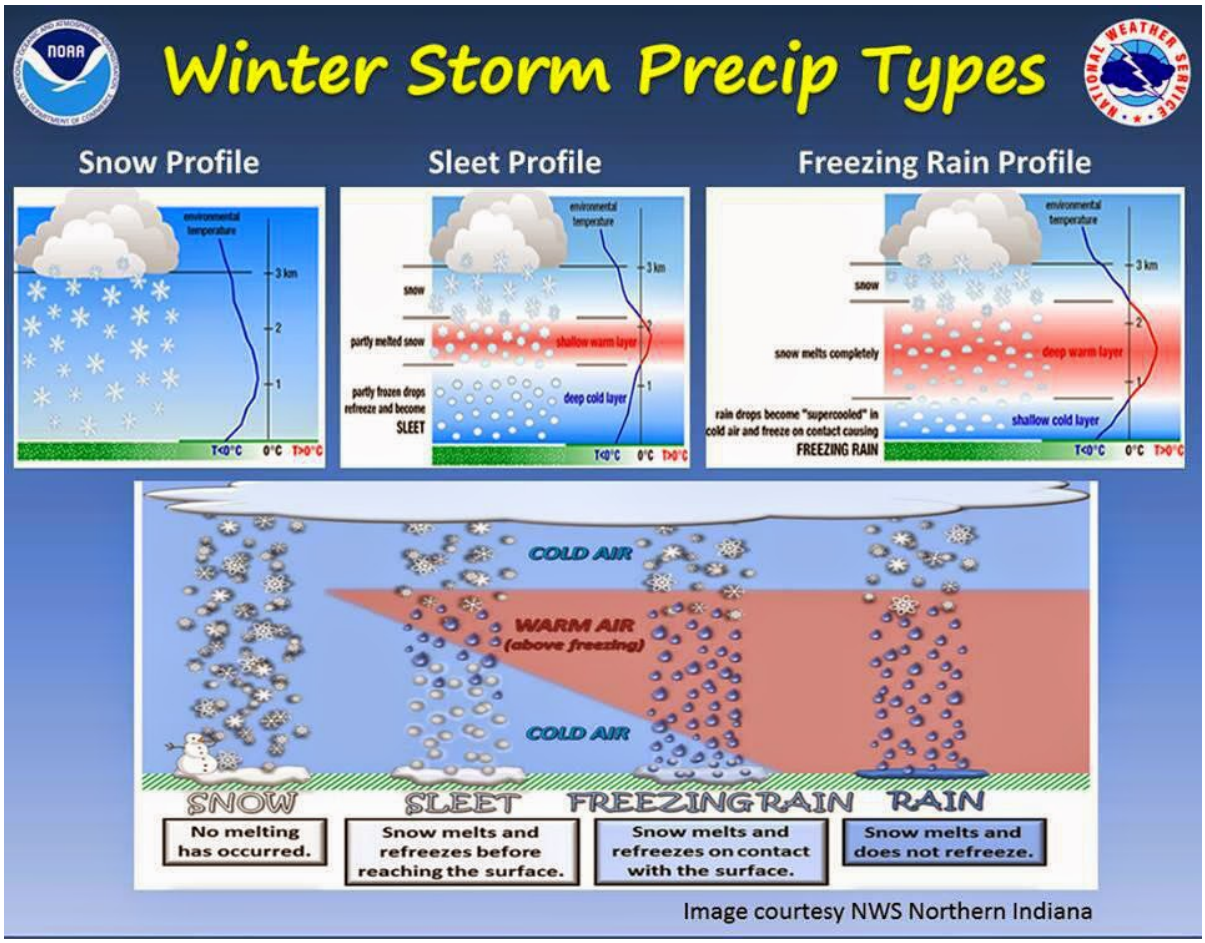
A. Supercooled droplets grow quickly because water vapor deposits preferentially on liquid water
B. Ice crystals grow at the expense of surrounding water droplets because water vapor condenses preferentially on ice
C. Precipitation rarely forms in mixed-phase clouds because it is difficult for air to become saturated with respect to both liquid water and ice
D. Ice crystals do not grow until all water droplets freeze because the saturation vapor pressure of ice is lower than that of liquid water

Solution: This is the Bergeron process, since the question is talking about precipitation forming in mixed-phase clouds. Recall that in the Bergeron process, ice crystals preferentially grow since vapor condenses on them more readily; this is also evidence as the saturation vapor pressure of ice is lower than liquid water, so it's easier to condense on ice than liquid water. So, B is the correct answer.

2.2 Types of Precipitation

Now that we know how precipitation particles form, we can look at what actually reaches the ground. The main factor that determines the type of precipitation is the **vertical temperature profile**, or how temperature changes with altitude.

Most precipitation in cold clouds begins as snow or ice. As it falls, it can pass through warm and cold layers that cause it to melt or refreeze. So, just knowing the temperature at the surface is not enough to determine what type of precipitation will occur.



How different temperature profiles produce snow, sleet, freezing rain, and rain (Source: National Weather Service)

The main types of precipitation are:

- **Rain**: Liquid water that reaches the ground. Rain can form directly through collision-coalescence, or snow can form higher in the cloud and melt as it falls through warmer air.
- **Drizzle**: Very small liquid droplets, usually less than around 0.5 mm in diameter. Drizzle commonly falls from low stratus clouds where there is not much vertical motion.
- **Snow**: Ice crystals that remain frozen as they fall to the ground. Snow does not necessarily require the entire atmosphere to be below freezing. Snowflakes can survive a shallow layer of air slightly above 0°C before completely melting.
- **Sleet**: Also known as **ice pellets**. Snow falls through a warm layer and melts into rain, but then enters a deep layer of subfreezing air near the ground. The droplets have enough time to freeze again before reaching the surface.
- **Freezing rain**: Snow first melts into rain in a warm layer, similar to sleet. However, the cold layer near the surface is too shallow for the droplets to refreeze before reaching the ground. Instead, they become **supercooled** and freeze when they hit roads, trees, power lines, and other surfaces.

The main difference between sleet and freezing rain is therefore the **depth of the cold layer near the surface**. A deeper cold layer allows the drops to freeze before reaching the ground, producing sleet. A shallow cold layer leaves the drops liquid until they hit the surface, producing freezing rain. The layer of ice produced by freezing rain is known as **glaze**.

Snow crystals themselves can also come in many different shapes. Temperature and humidity determine whether crystals grow as plates, columns, needles, or branching **dendrites**. Snowflakes are often not single crystals either, but groups of crystals that have stuck together through aggregation.

There are a few other types of precipitation that are especially important for storms:

- **Graupel**: Forms when supercooled droplets freeze onto a snowflake or ice crystal through riming. Graupel is white, soft, and easier to crush than sleet.
- **Hail**: Large balls or lumps of ice that form inside thunderstorms. Hail requires strong updrafts and lots of supercooled water.
- **Virga**: Precipitation that falls from a cloud but evaporates or sublimates before reaching the ground.

Let's look more closely at hail. A hailstone usually begins as a small ice particle or piece of graupel. Strong thunderstorm updrafts carry this particle through regions containing supercooled water. These droplets collide with the particle and freeze onto it, causing the hailstone to grow.

The stronger the updraft, the longer a hailstone can remain suspended inside the cloud. This gives it more time to collect water and grow larger. Eventually, the hailstone becomes too heavy for the updraft to support, moves outside of the strongest part of the updraft, or falls after the updraft weakens. This is why **supercells**, which have very strong and long-lasting updrafts, can produce the largest hail.

Hailstones can also have alternating clear and cloudy layers. If water freezes slowly, air bubbles can escape and the ice is relatively clear. If the water freezes quickly, air bubbles become trapped and create cloudy ice.

Lastly, **virga** occurs when precipitation falls into a dry layer beneath a cloud and evaporates or sublimates before it reaches the surface. Virga often looks like streaks hanging below a cloud that disappear before reaching the ground.

Virga is also important for thunderstorms because evaporation cools the surrounding air. Cooler air is denser, so it begins to sink and can strengthen a **downdraft**. We will talk more about downdrafts and their role in thunderstorms later in this handout.

> **USESO Open 2026, Section I Question 26** Cold air damming occurs when cold air gets trapped by easterly winds along the windward side of a mountain range. Which of the following statements is/are true about cold air damming configurations in the Northern Hemisphere?

I) Cold air damming configurations typically have high pressure to the south
II) Cold air damming is associated with freezing rain

A. I only B. II only C. I and II D. None

Solution: High pressure to the south would create a southwesterly wind, moving air away from the mountain—I is false. In a cold air damming setup, the warmer air overruns the colder air, producing drizzle that becomes supercooled upon entering the colder air below—II is true.

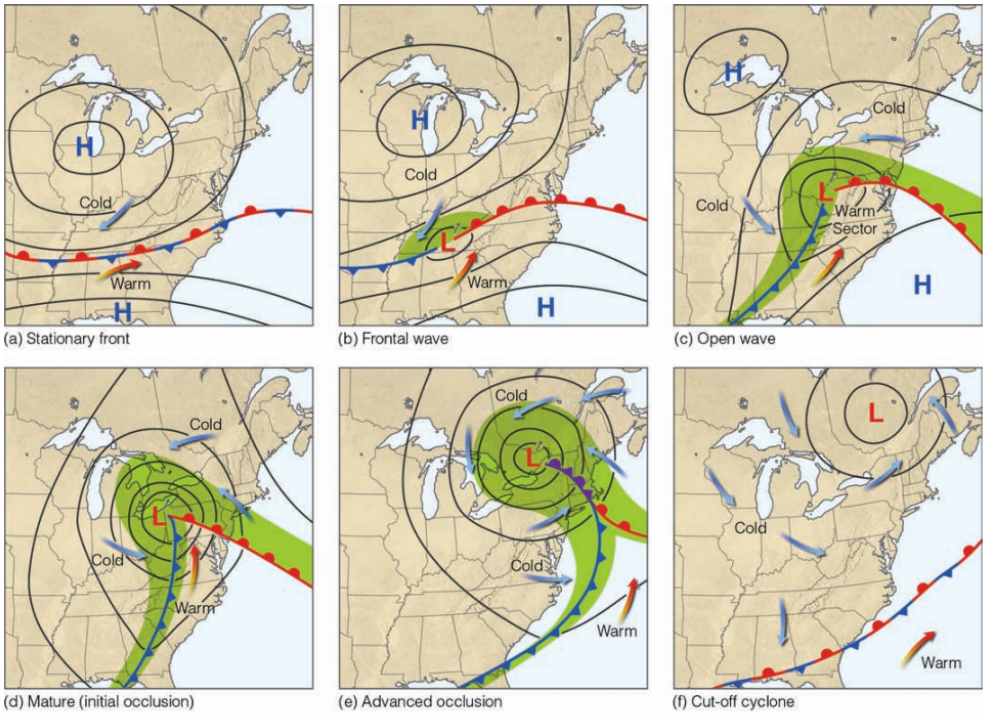
3 Extratropical Cyclones

Extratropical cyclones are some pretty powerful storms. They are also known as **wave cyclones** or **mid-latitude cyclones**. They drive most of the weather that occurs outside the tropics, between roughly 30 and 60 degrees latitude north and south. This weather can include all the types of precipitation mentioned above, as well as thunderstorms and tornadoes.

3.1 Formation

As you already know from the Pressure handout, cyclonic rotation occurs around a region of low pressure. The polar front, a stationary front, is a trough of low pressure, due to the action of the Ferrel and Polar cells. The Coriolis force then causes geostrophic winds to blow eastwards south of the stationary front and westwards north of the front. This creates a cold front and a warm front next to each other, which move cyclonically counterclockwise around the central low. Cold fronts move faster than warm fronts, so eventually an occluded front will form, starting closest to the central low and eventually moving its way outwards. Finally, the occlusion ends, cutting off the cyclone into an isolated low pressure system while the stationary front reforms.

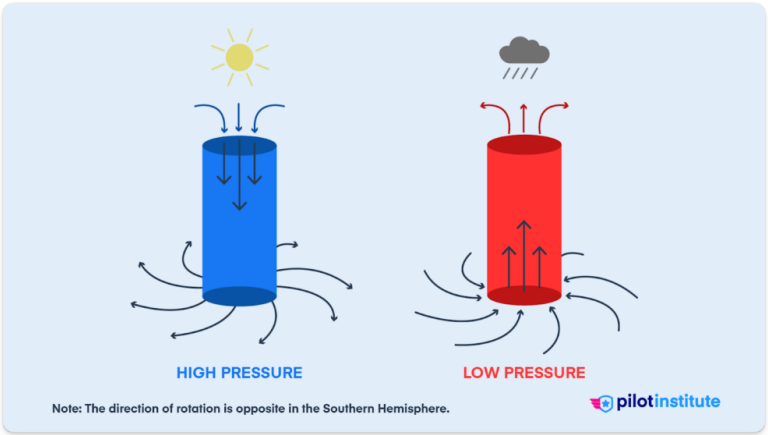
This overall process is known as **cyclogenesis**. A **nor'easter** is a type of extratropical cyclone that forms along the East Coast of the United States.



● FIGURE 12.1 The idealized life cycle of a mid-latitude cyclone (a through f) in the Northern Hemisphere based on the polar front theory. As the life cycle progresses, the system moves northeastward in a dynamic fashion. The small arrow next to each L shows the direction of storm movement.

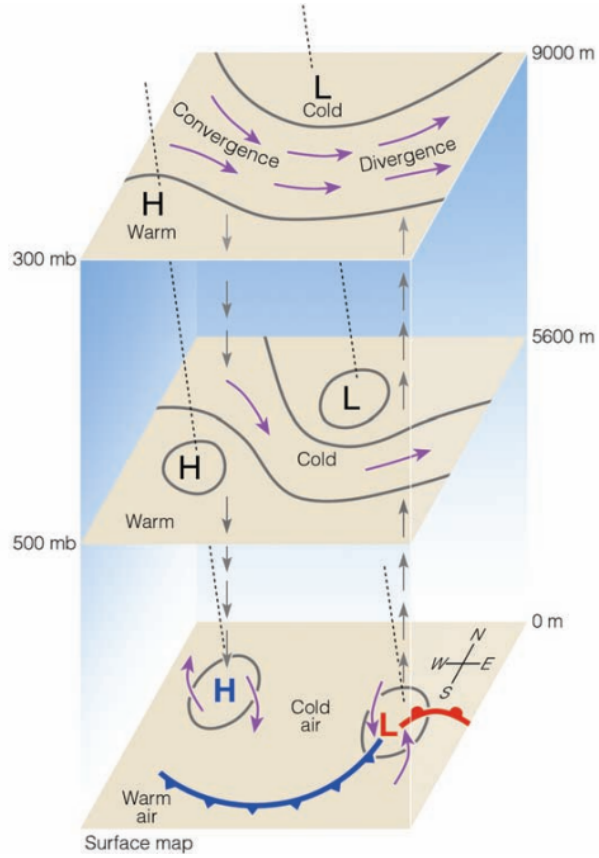
The evolution of a single extratropical cyclone (Source: Ahrens)

3.1.1 Jet Stream Involvement



Air moves upwards from a low pressure system, such as an extratropical cyclone. (Source: Pilot Institute)

Upper level wind flows, such as the jet stream, are also very important in the formation of an extratropical cyclone. Since these storms are a low pressure system, winds blow inward, which means there must be a net transport of air upwards towards the jet stream. So, the jet stream must diverge to allow for more air to flow away than towards the spot above the storm.



● FIGURE 12.7 An idealized vertical structure of a middle-latitude cyclone and anticyclone.

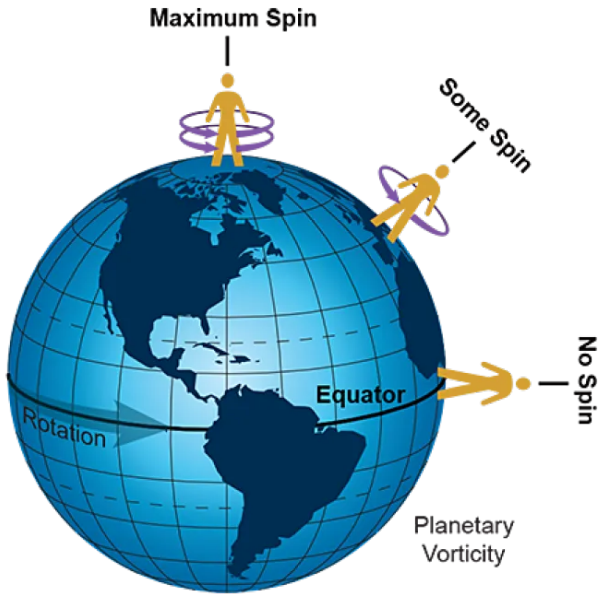
Vertical structure of an extratropical cyclone (Source: Ahrens)

3.2 Vorticity

Vorticity is a common subject of USESO Camp Exam questions. Here, we will give a qualitative, non-mathematical introduction to the subject, as the linear algebra can get quite complicated.

Vorticity measures the spinning or rotational motion of an object, such as an air parcel. There are two components:

- **Planetary vorticity** occurs due to Earth's rotation. It is represented with the letter f. Since an object doesn't rotate at the equator, $f = 0$ at the equator. An object's rotation is maximized at the poles, so f has the highest magnitude at the poles: Negative in the Southern Hemisphere and positive in the Northern.
- **Relative vorticity** occurs due to an object spinning relative to Earth. It is represented with the Greek letter zeta: ζ. We define cyclonic rotation to have positive vorticity and anticyclonic rotation to have negative vorticity.



Planetary Vorticity (Source: NOAA)

Finally, the Greek letter η [1] denotes the **absolute vorticity,** which is a conserved quantity. This means that if the planetary vorticity increases, the relative vorticity must decrease.

Due to the conservation of vorticity in the jet stream, as it moves from a ridge to a trough, towards the equator, f decreases, so ζ must increase. Vice versa, as air moves from a trough to a ridge, toward the poles, ζ decreases. This transfer of relative vorticity ζ is known as **vorticity advection**.

When relative vorticity increases (positive vorticity advection) as air moves poleward, atmospheric waves known as shortwaves cause the jet stream to diverge (see the image in the section above), which encourages the formation of cyclones.

USESO Camp Exam 2025, Section II Problem 2.2(b) Recall that the absolute vorticity ν (derived from column rotation), planetary vorticity f (derived from Earth's rotation), and relative vorticity ζ are related by $\nu = f + \zeta$. Winds from A are advecting _____ values of f and _____ values of ζ towards B, assuming constant ν.

A. Higher, higher
B. Higher, lower
C. Lower, higher
D. Lower, lower

[1] Sometimes, ν (nu) is used instead.



Solution: Since we are moving closer to the equator, f is decreasing. So, vorticity advection is moving higher vorticities from A to B. Since f decreases and ν is constant, ζ must increase, so lower values are advected. So, the correct answer is A.

4 Thunderstorms

The second type of storm we will discuss is the **thunderstorm**. These storms can range from mild to severe, generally depending on what type of storm they are: **ordinary thunderstorms** tend to be weaker, while **multicells** and **supercells** can cause widespread destruction through flooding and strong winds.

4.1 Formation

The different types of thunderstorms all require some similar factors, listed below:

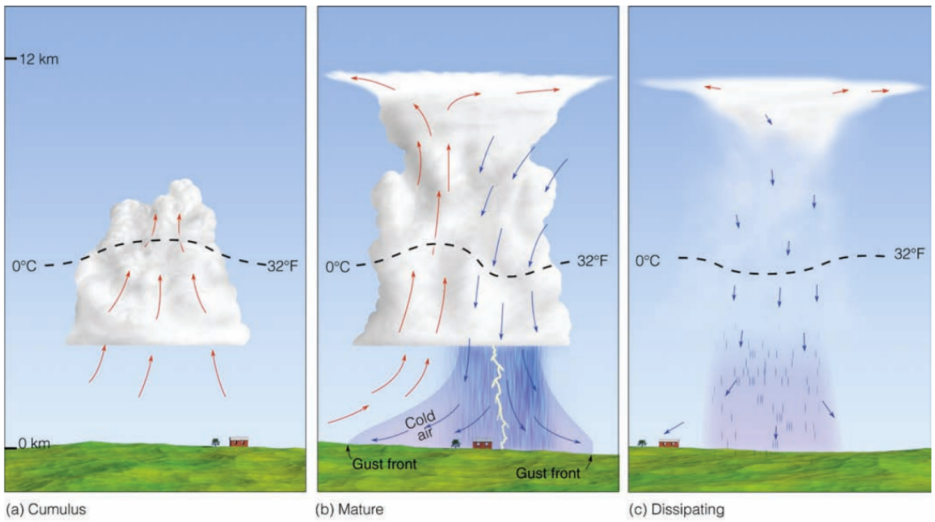
- Warm, moist air
- Conditionally unstable atmosphere (moist air rises)
- A **lifting mechanism**
- Optional: Vertical shear can help thunderstorms last longer

Lifting mechanisms are triggers that lift air to the lifting condensation level (LCL), where it reaches its dew point, becomes saturated, and rises on its own in the conditionally unstable atmosphere. These mechanisms include:

- **Convective lifting**: Air is heated by the Sun, warm air rises
- **Orographic lifting**: Terrain/mountains force air upwards
- **Frontal lifting**: Warm air moves over a front
- **Convergence lifting**: Air converging has no where to go but up

In an ordinary thunderstorm, the lifting mechanism is usually convective. A cumulus cloud forms as air is uplifted. Then, through the collision-coalescence process, rain starts to fall, sometimes accompanied with thunder. This leads to a downdraft, which eventually cuts off new moist air, dissipating the storm.

Additionally, dry air from around the cloud will be drawn into the cloud, which decreases moisture levels in the cloud, impairing the formation of more rain. This process is known as **entrainment**. Lastly, due to the temperature inversion in the stratosphere, updrafts cannot extend past the tropopause, which leads to the cloud tops flattening out into an **anvil cloud**.



● FIGURE 14.2 Simplified model depicting the life cycle of an ordinary cell thunderstorm that is nearly stationary as it forms in a region of low wind shear. (Arrows show vertical air currents. Dashed line represents freezing level, 0°C isotherm.)

Life cycle of an ordinary thunderstorm (Source: Ahrens)

These thunderstorms are usually short-lived, lasting at most a few hours. Since there is limited vertical wind shear, the updraft can't sustain itself when the downdraft appears. However, in the next subsection, the updraft and downdraft can exist simultaneously, leading to more intense storms.

4.2 Multicells and Supercells

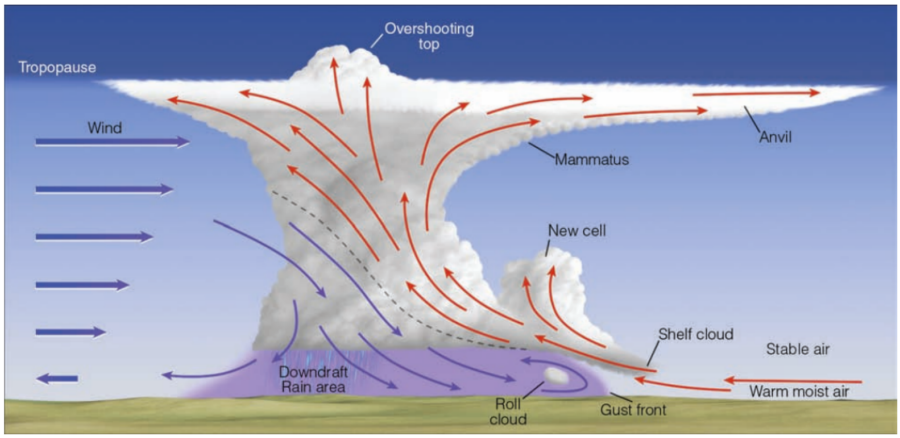
4.2.1 Multicells

Now, we will talk about multicell thunderstorms. These thunderstorms are formed when there is strong vertical speed shear, which means that the winds move the same direction at the surface and aloft, but the speed is faster the higher you go. This allows for the updraft to "ride over" the downdraft, allowing both to coexist at the same time. This leads to a much longer lasting storm. These storms are called multicells because updrafts can form a new cell of the thunderstorm instead of just feeding back into the original circulation cycle.

Components:

- **Overshooting top**: Updrafts can force air through the tropopause into the stratosphere, forming a part of the cumulonimbus cloud that extends above the anvil.
- **Mesohigh**: A small, shallow high pressure area below the downdraft.

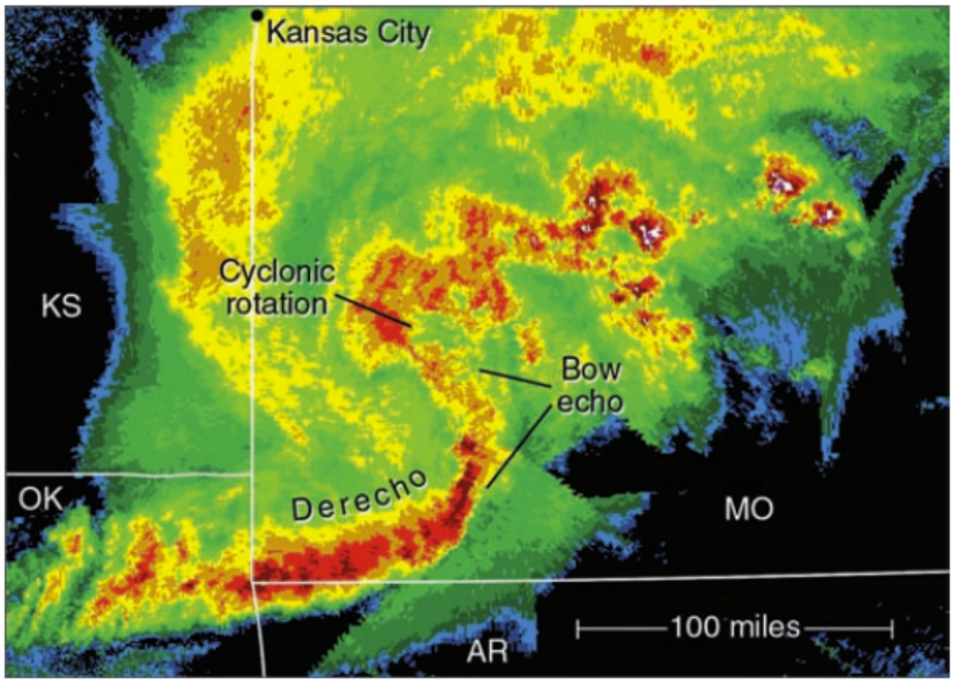
- **Gust front**: Boundary between the downdraft and still air. Has strong winds that flow in a straight line outwards, known as **straight line winds**. When multiple gust fronts converge, air is forced upward, leading to the formation of new thunderstorms. Also known as an outflow boundary.
- Shelf/arcus cloud: Forms along the front edge of the gust front.
- Roll cloud: Detached cylindrical cloud above the gust front.
- Mammatus clouds can form downward from an anvil cloud.



● FIGURE 14.5 A simplified model describing air motions and other features associated with an intense multicell thunderstorm that has a tilted updraft. The severity depends on the intensity of the storm's circulation pattern.

Anatomy of a multicell (Source: Ahrens)

Squall lines are organized rows of thunderstorms, and these storms are often multicells. These rows sometimes form as a result of a large front creating the same conditions across a wide area. This front is often created as the result of an extratropical cyclone. Due to cyclonic rotation, squall lines form a **mesoscale convective vortex** near the northern end and move into a curve known as a **bow echo** on Doppler radar. A **derecho** is a strong straight-line wind originating out of the bow echo. By definition, these winds exceed 50 knots (58 mph), and can reach over 80 knots in extreme cases.

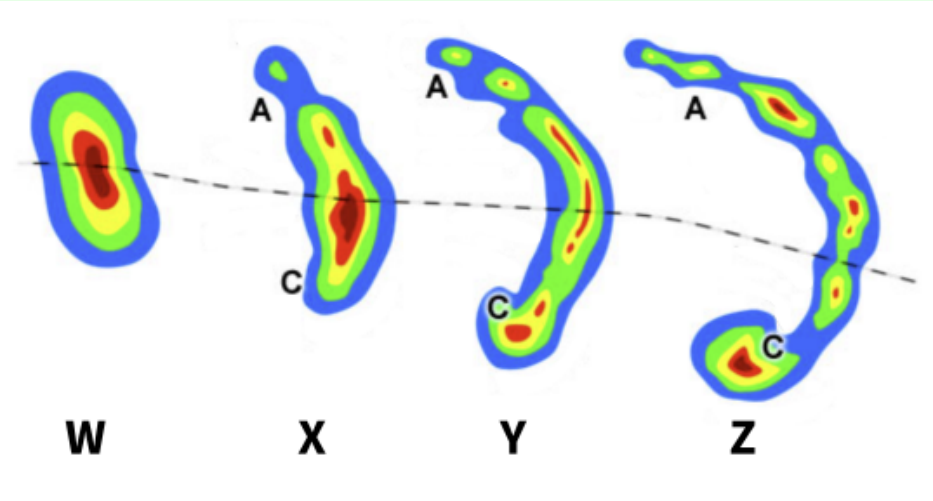


● **FIGURE 14.17** A Doppler radar image showing an intense squall line in the shape of a bow—called a *bow echo*—moving eastward across Missouri on the morning of May 8, 2009. The strong thunderstorms (red and orange in the image) are producing damaging straight-line winds over a wide area. A region of straight-line winds that extends for a good distance along a squall line is called a *derecho*.

A squall line that has moved into a curve, registering as a bow echo on Doppler radar. Cyclonic rotation is observed on the northern end of the squall line. (Source: Ahrens)

Mesoscale convective systems (MCSs) are complexes of organized thunderstorms that persist for hours or more. Squall lines are a type of MCS. **Mesoscale convective complexes (MCCs)** are a type of large MCS that can cover entire states.

USESO Camp 2025, Section I Question 28 Certain arrangements of thunderstorms can produce a radar signature called a bow echo, which can morph into a comma echo over time. The figure below depicts a complex of thunderstorms (W) evolving into a bow echo (X and Y) and then into a comma echo (Z). The red shading indicates where the storm is most intense.



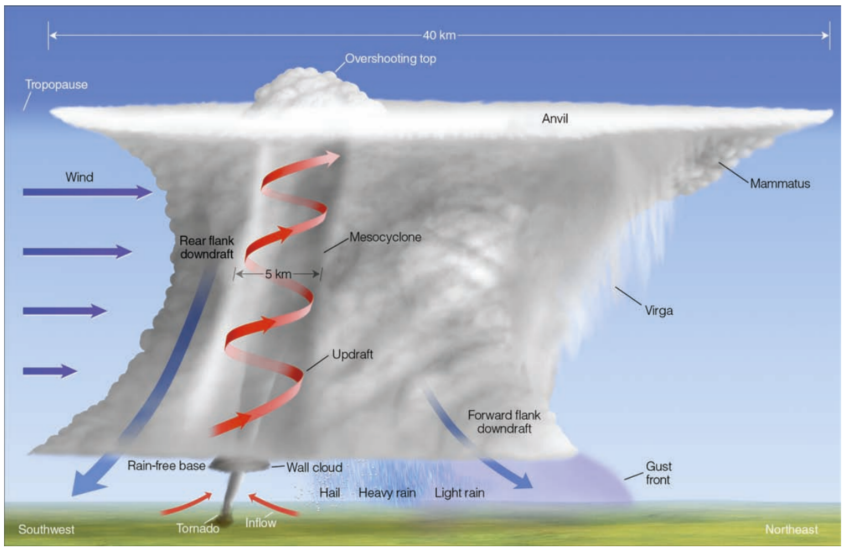
Which of the following accurately describes the evolution of the bow echo and the hemisphere in which the storm occurred?

A. Cyclonic winds at A are weakened while anticyclonic winds at C are strengthened; Northern Hemisphere
B. Anticyclonic winds at A are weakened while cyclonic winds at C are strengthened; Northern Hemisphere
C. Cyclonic winds at A are weakened while anticyclonic winds at C are strengthened; Southern Hemisphere
D. Anticyclonic winds at A are weakened while cyclonic winds at C are strengthened; Southern Hemisphere

Solution: The storm is stronger at C, since it is shaded red. So, we would expect it to be cyclonic. Additionally, we know from the image above that in an MCS, cyclonic rotation is present closer to the poles. So, we are in the Southern Hemisphere, as the cyclonic system is farther south. So, the correct answer is D.

4.2.2 Supercells

Supercells are thunderstorms with a rotating updraft called a **mesocyclone**. These storms form with strong speed or directional vertical wind shear, which means the strength and direction of the wind vary as you increase in altitude. This updraft allows the cloud to exist for hours. It is also strong enough that hail can continue to be blown upwards until it grows very large, so supercell thunderstorms form the largest hail.



● FIGURE 14.23 Some of the features associated with a classic tornado-breeding supercell thunderstorm as viewed from the southeast. The storm is moving to the northeast.

How a mesocyclone forms (Source: Ahrens)

Components:

- Shares anvil/mammatus clouds, gust fronts, and overshooting tops with multicell thunderstorms
- Mesocyclone (see above)
- **Rain-free base**: There is no rain under the mesocyclone.
- Wall cloud: Rotating, descending cloud under the mesocyclone.
- Hail, rain and virga fall from supercells. Virga is rain that evaporates before reaching the ground.

Supercells form in an interesting way. A stable temperature inversion can "cap" a cloud from expanding upwards further, which prevents small thunderstorms from forming. So, heat and moisture accumulate at the surface. The quantity of heat and moisture energy is measured by the **Convective Available Potential Energy (CAPE).** If enough CAPE is available and a strong lifting mechanism forces air upwards, the updraft can burst through the temperature inversion, leading to explosive growth and eventually a supercell thunderstorm. These conditions are often found in the open wave stage of extratropical cyclone formation.

The formation of a single supercell inhibits the formation of others, since it steals a lot of the necessary moisture supply. So, most supercells are isolated.

4.3 Hazards

Many natural hazards are associated with thunderstorms. These can be both economically damaging as well as dangerous to humans.

4.3.1 Flooding

Flooding can happen if a lot of rain falls on a location in a short time interval. This can occur when a thunderstorm stalls (stays in the same location), moves very slowly, or trains(moves over the same location many times). A **flash flood** is a type of flood that develops rapidly with little or no advance warning. These can be very dangerous to people and property, especially in low elevation regions.

4.3.2 Wind and Turbulence

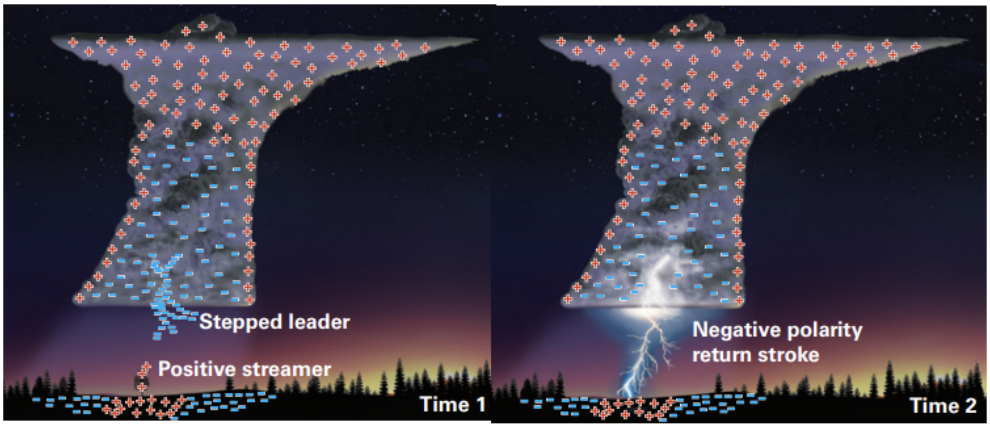
Turbulence is the violent or unsteady movement of air in the atmosphere. It is often associated with thunderstorms, which destabilize the atmosphere. Along with strong winds, it makes thunderstorms quite dangerous, especially for aircraft:

- At gust fronts, straight-line winds and derechos can be strong enough to blow down houses and trees.
- **Clear air turbulence (CAT)** is a type of turbulence that is invisible, and occurs in clear air with no clouds. It occurs when upper-level air interacts with the anvil cloud. A very dangerous type of CAT is the **microburst,** a short-lived, intense downdraft, which hits the ground and spreads out and upwards. This violent movement can strip away lift and crash planes.

4.3.3 Lightning

Lightning is the discharge of electricity from a cloud to either another cloud or the ground. **Thunder** is the associated sound wave. Thunder travels slower than lightning. After you see a lightning strike, you can count the number of seconds it takes for you to hear the thunder. Divide that by five, and that's the distance to the thunderstorm in miles.

Lightning forms when clouds gain charge, typically positive near the ground and outside of the cloud and negative near the inside of the cloud. This occurs when rain and hail move through the cloud on updrafts and downdrafts while transporting charges on their surfaces. Eventually, the charge difference will be great enough that the insulating air breaks down and becomes a conductor, transferring charges rapidly.



(c) The formation of a cloud-to-ground stroke takes place in stages, over the course of microseconds.

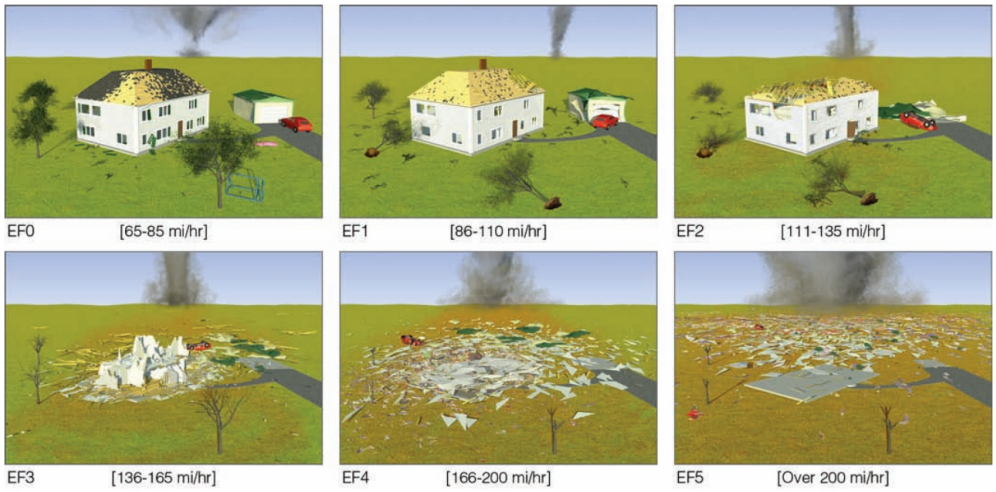
Charges being transferred during a lightning strike (Source: Marshak)

4.3.4 Tornadoes

A **tornado** is a rapidly rotating column of air extending down from a cumuliform cloud. It has wind speeds between 100 and 220 knots, and range in width from a few meters to a mile. Generally, tornadoes form under supercell thunderstorms. They are common in the American Midwest, along Tornado Alley between Texas and Nebraska, as well as Dixie Alley, in Mississippi and Alabama.

Alerts about tornadoes mainly fall into two categories: watches and warnings. A **tornado watch** is given when the atmosphere has the right conditions for a tornado to form. A **tornado warning** is given when a tornado is about to form.

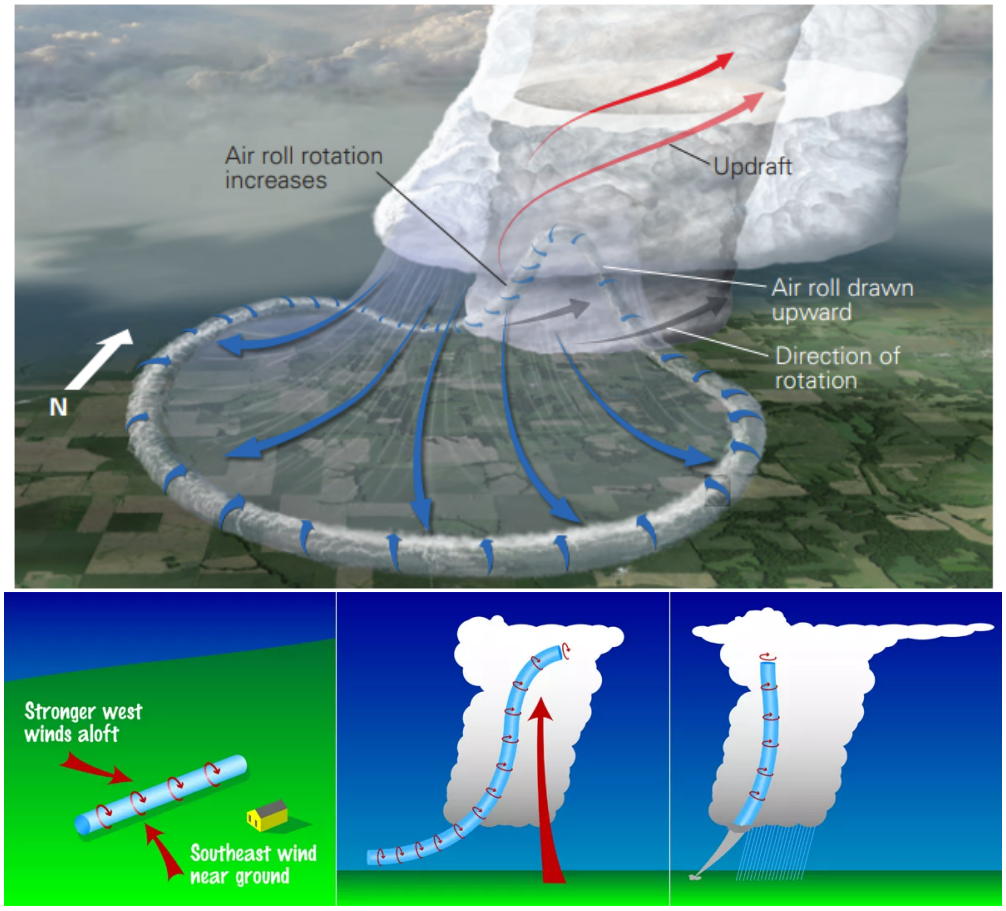
Tornadoes are classified on the **Enhanced Fujita scale,** which measures the destruction caused by a tornado, which correlates with wind speeds.



● FIGURE 15.11 Damage to the house in Fig. 15.10 and its surroundings caused by tornadoes of varying EF intensity.

The Enhanced Fujita Scale (Source: Ahrens)

Tornadoes form when a downdraft along the forward or rear flank of a supercell thunderstorm creates a horizontal roll of air known as a **vortex tube**. Updrafts then tilt this tube, as it gets pulled into the mesocyclone. Eventually, a visible funnel of dust, debris, and cloud particles—the tornado—will be formed.

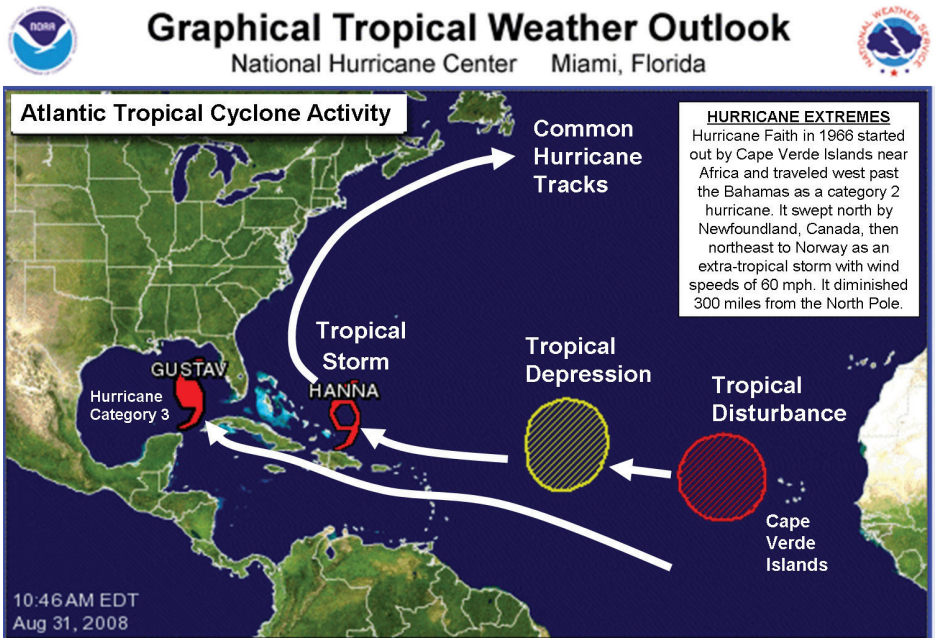


Formation of a tornado (Source: Marshak(top) and NOAA NESDIS(bottom))

5 Hurricanes

The last type of storm we will discuss are **hurricanes**. You've probably heard about hurricanes, but have you ever heard of cyclones or typhoons? All of these describe the same phenomenon, which we will more generally refer to as **tropical cyclones (TCs)**.

Hurricanes are prevalent throughout the world, and they are associated with many destructive hazards. Usually, hurricanes occur during summer months due to prevalence of warm water which powers the hurricane. We will start by looking at the life cycle of hurricanes.



Atlantic TC Tracks (Source: NOAA National hurricane center)

5.1 Life Cycle

5.1.1 Formation and Intensification

Tropical cyclones start off in the tropics where warm water and heat are abundant with other factors also important for formation. Factors favorable for formation:

- Abundant warm water (both widespread on the surface and in depth) and solar insolation
- Low wind shear
- High moisture
- Pre-existing Disturbance
- Nonzero Coriolis force (not directly on equator: $> 5°$ N/S usually)

Usually if there's a preexisting low pressure disturbance from another event, this forces a **positive feedback loop** as low pressure regions draw more air in, and these spiraling winds pick up heat and moisture from evaporating seawater. And this warm moist air is lighter than the surrounding air causing convection, and eventually the moisture in the air condenses, releasing latent heat which further lowers surface pressure - amplifying the feedback. This feedback loop is also known as **Wind-induced surface heat exchange (WISHE)**.

1. Beginning - Tropical cyclones usually start off as **tropical waves** (an elongated trough of low pressure moving westward due to easterlies), but they can also can start off in other ways as well like frontal boundaries and thunderstorm clusters.
2. Clustering - They can then intensify and loosely organize themselves into **tropical disturbances** which are loose, disorganized groups of thunderstorms.

3. Organization - When these thunderstorms organize around a center of rotation, a **tropical depression** forms ($< 33kts$). Vague rainbands form.

4. Tighter organization and circulation - **tropical storms** ($34 - 63kts$); the rainbands become more distinctive and storms get a name now

5. More intensification - **Hurricane/Typhoon status** ($64 + kts$).; defined clear structure, prominent rain bands, and clear eye forms

Different tropical cyclone/hurricane intensities are measured on the **Saffir-Simpson scale.**

(Note: A lot of the storms on the atlantic form due to tropical waves coming in the summer from Africa, called the African easterly waves which are generated by the **African Easterly Jet's** instability.)

SAFFIR-SIMPSON SCALE

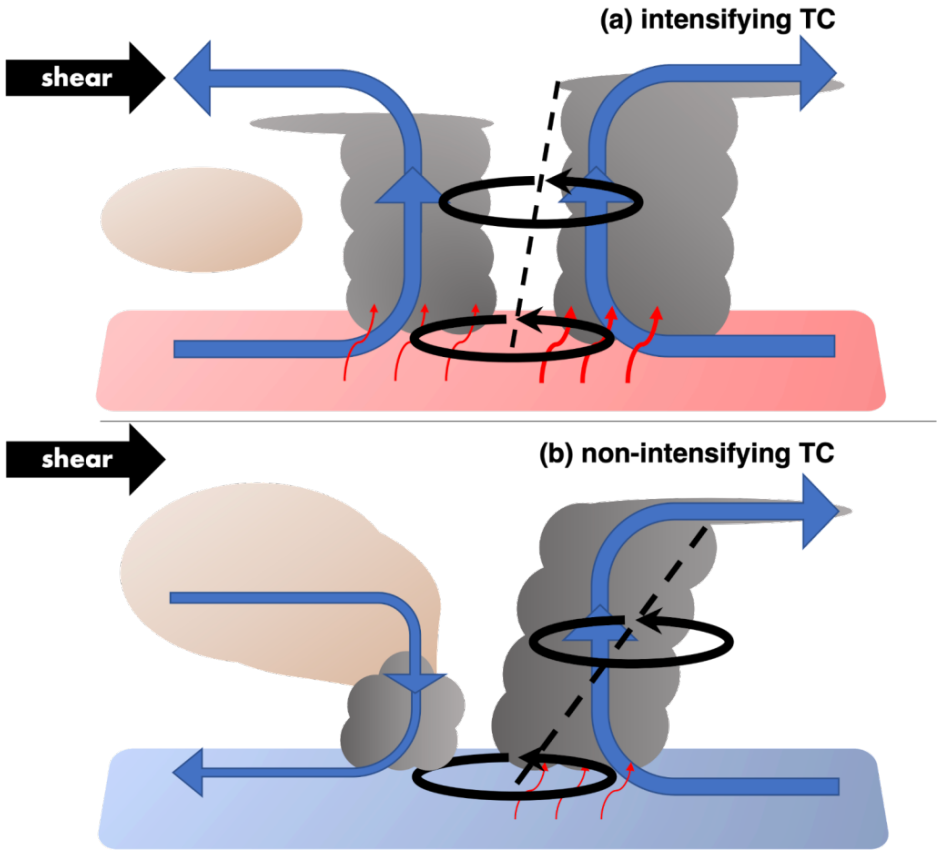
STRENGTH	WIND SPEED MPH	WIND SPEED KT	DAMAGE
CATEGORY 1	74-95 MPH	64-82 KT	SOME
CATEGORY 2	96-110 MPH	83-95 KT	EXTENSIVE
CATEGORY 3	111-129 MPH	96-112 KT	DEVASTATING
CATEGORY 4	130-156 MPH	113-136 KT	CATASTROPHIC
CATEGORY 5	157+ MPH	137+ KT	CATASTROPHIC

Saffir-Simpson Scale (Source: NOAA)

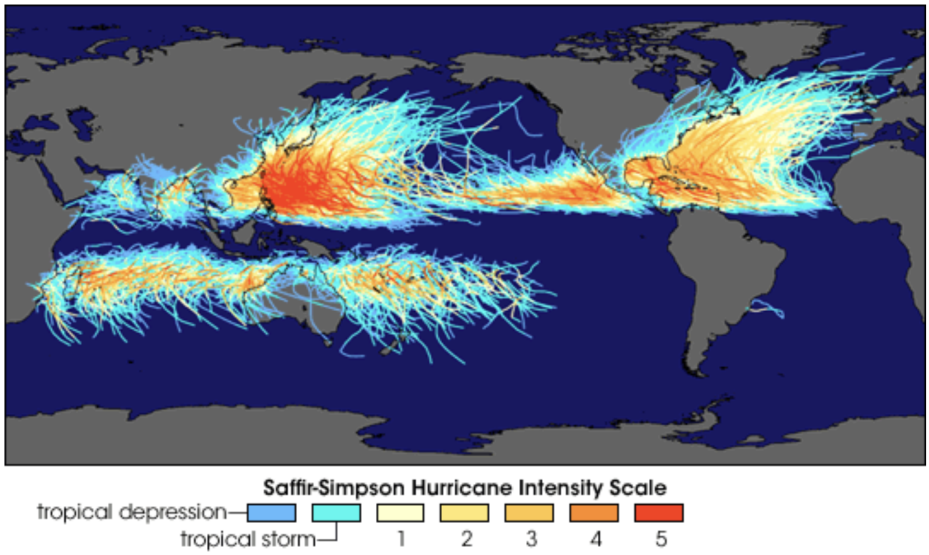
For intensification, multiple parameters matter like shear, how dry the air above is (affects ventilation, amount of dry air that makes its way into the storm's inner core (eye + eyewall), humidity, the depth of the warm water in the ocean (as cold water right below can be surfaced by wind pushing away warm water and cause a **negative feedback** slowing down intensification) along with many other parameters.

5.1.2 Landfall/Decay

Hurricanes can decay for multiple reasons with the main one being their fuel being cutoff. This can happen through landfall as there's not as much warm water to gain energy from in the form of **latent heat**. In addition, traveling over a colder patch where evaporation is more difficult can cause decay. However, there are other methods like wind shear causing the storm to tilt, meaning that the low pressure region loses energy and gains pressure by allowing dry air to come in and destroy the low pressure region (**Ventilation**).



Shear and Ventilation causing weakening of the storm (Source: NOAA)

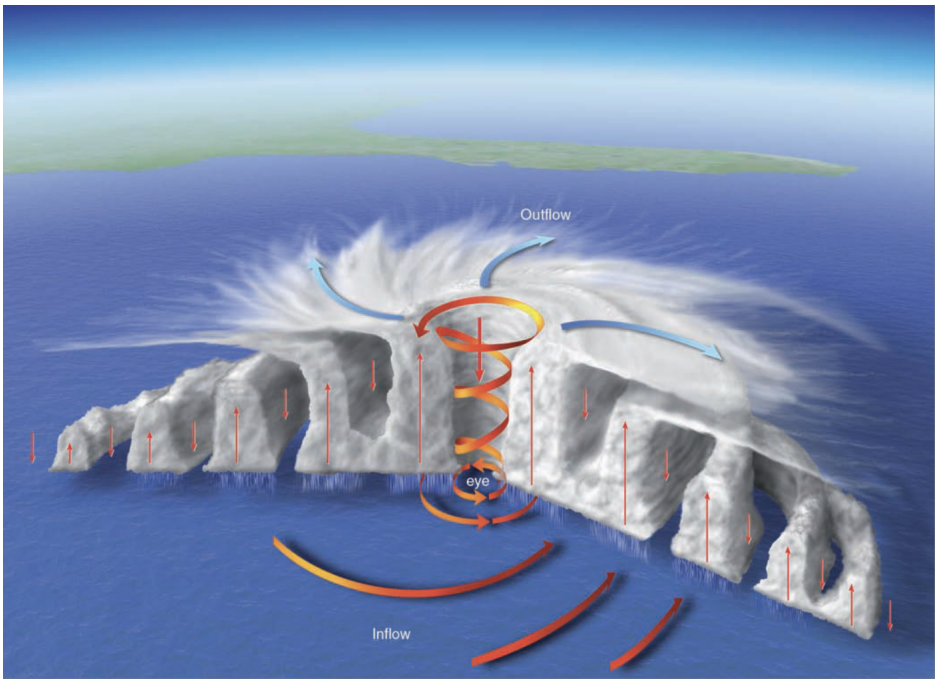


World TC Tracks (Source: NASA Science)

5.2 Anatomy

5.2.1 General Structure

These events are **cyclonic**, meaning that they have a **low-pressure center**. This means that they spin counterclockwise in the Northern Hemisphere and clockwise in the Southern Hemisphere.



Tropical Cyclone Structure (Source: Ahrens)

5.2.2 The Eye

The eye of the storm is the low-pressure center and is usually clear of clouds as air sinks slowly from upper-levels causing drying air. This area has very low wind speeds compared to the eyewall right outside. It is also usually the warmest part of the storm.

5.2.3 Eyewall

This is where the highest amount of convergence and rising of air occurs (convection), and the winds are the most intense here. The eyewall is not actually a vertical cylinder but it actually leans outward with height because of spin decreasing with height and **conservation of angular momentum**.

Hurricanes have a lot of variability with their radius and winds. Phenomena like **eyewall replacement cycles** cause intensification to vary, and eyewalls can have many different **polygonal shapes** like hexagons.

5.2.4 Spiral Rainbands

These are curved bands of clouds (convection occurs) extending outward from the eyewall and often have intense rainfall and sometimes even tornadoes. These are often discontinuous with dry sinking air in between these rainbands. They propagate radially outward due to effects with vortex rossby waves. Note: There are many different types of rainbands.

5.2.5 Upper Atmosphere - The Outflow Region

The outflow region's job is **mass removal**, and to balance convergence at the surface, divergence must occur aloft.

Why the top is anticyclonic. Pressure falls more slowly with height in warm air, so a hurricane's warm core flips into a high near the tropopause, which circulates anticyclonically. (Note: this is separate from outflow itself. Outflow is radial motion, anticyclonic is rotation, and air can flow outward while still spinning cyclonically and often does.)

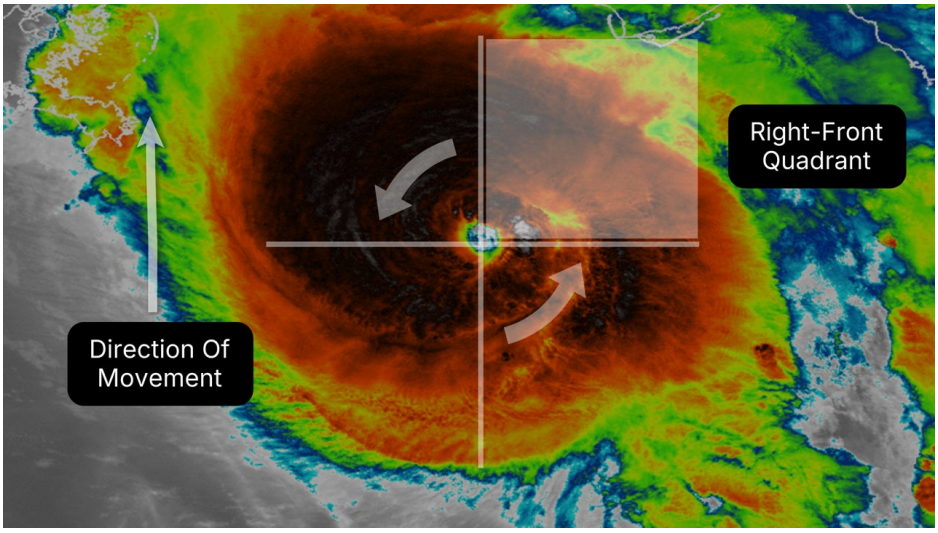
Cirrus clouds. The clouds we associate with hurricanes are **cumulonimbus** towers with bases near the lifting condensation level (∼500 m). By 12-16 km the air has dumped its water as rain and released its heat, and anything left is ice. The bright dense area over the core is the **central dense overcast**.

Why satellite images still look cyclonic. The spiral bands you see are cloud tops anchored to the cyclonic circulation below, not a picture of the upper-level wind. The flow also stays cyclonic out to a couple hundred km before reversing, since anticyclones cannot support strong winds near their axis and are only dynamically permitted at large radius.

Outflow jets. Outflow is rarely a symmetric fan. It usually concentrates into one or two narrow channels of fast air called **outflow jets**, visible on satellite as cirrus plumes streaming off one side of the storm and curving anticyclonically. They form wherever the surrounding upper-level flow offers the least resistance, measured by **inertial stability**, which is lower at lower latitudes due to a smaller coriolis parameter, so the equatorward side is often favored. Poleward channels are also common and typically form when an upper-level trough approaches and couples with the outflow. **Dual-channel outflow** is a strongly favorable signature for rapid intensification.

5.2.6 Quadrants

There are 4 quadrants that we can divide a tropical cyclone to, and depending on what quadrant you may be hit by, the outcomes can be drastically different. (Note: Currently, we are assuming Northern Hemisphere, but in the southern hemisphere, just reverse the right to left as the storms are spinning the opposite direction. Also, the front quadrant is defined as the one on the side toward the direction of translation of the tropical cyclone).



Tropical Cyclone Quadrants (Source: Weather Channel)

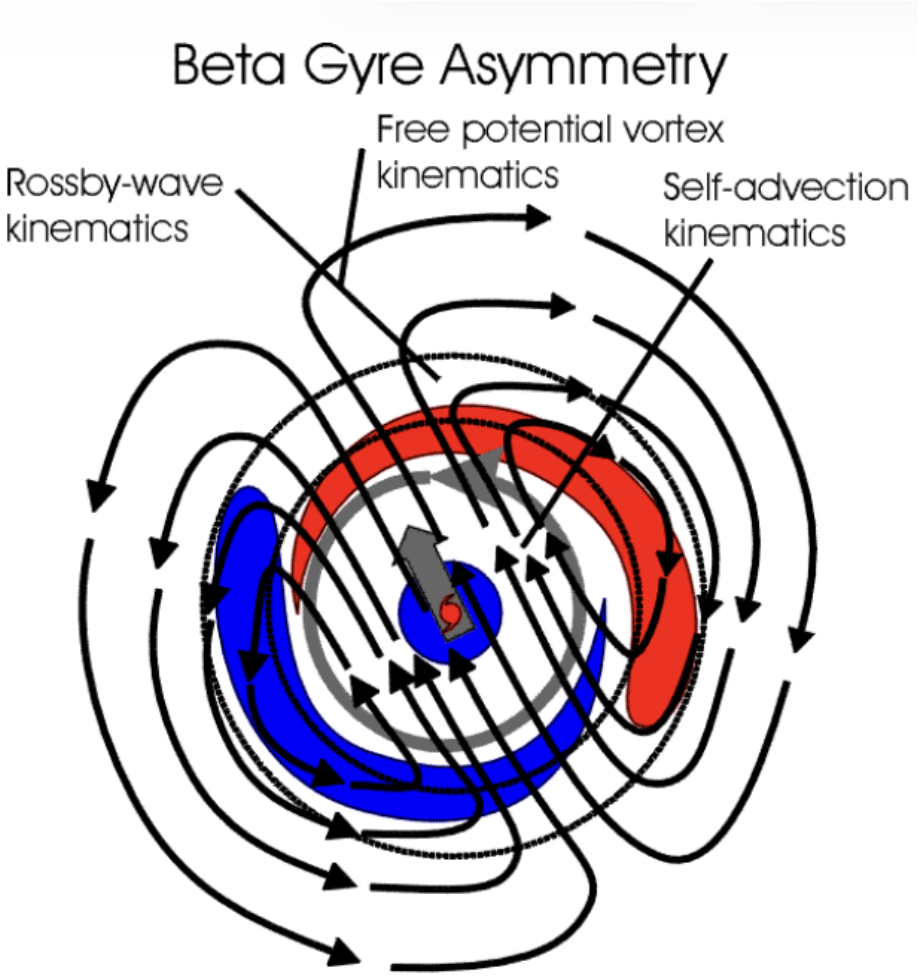
Tropical cyclones both spin and translate, meaning that when they can add either constructively or destructively. This means that in the northern hemisphere, the right side has translation that is aligned with it's rotation and vice versa for the left side. In addition, the rear quadrants have less energy usually as their energy is derived by water that has already lost energy to the front

quadrants. This means, if you're in the direction of landfall of the right-front quadrant, you may be expected to have fast winds, high storms surge, and a higher tornado threat.

Note: Size of a storm and intensity of the storm aren't necessarily related as intense storms, more independent.

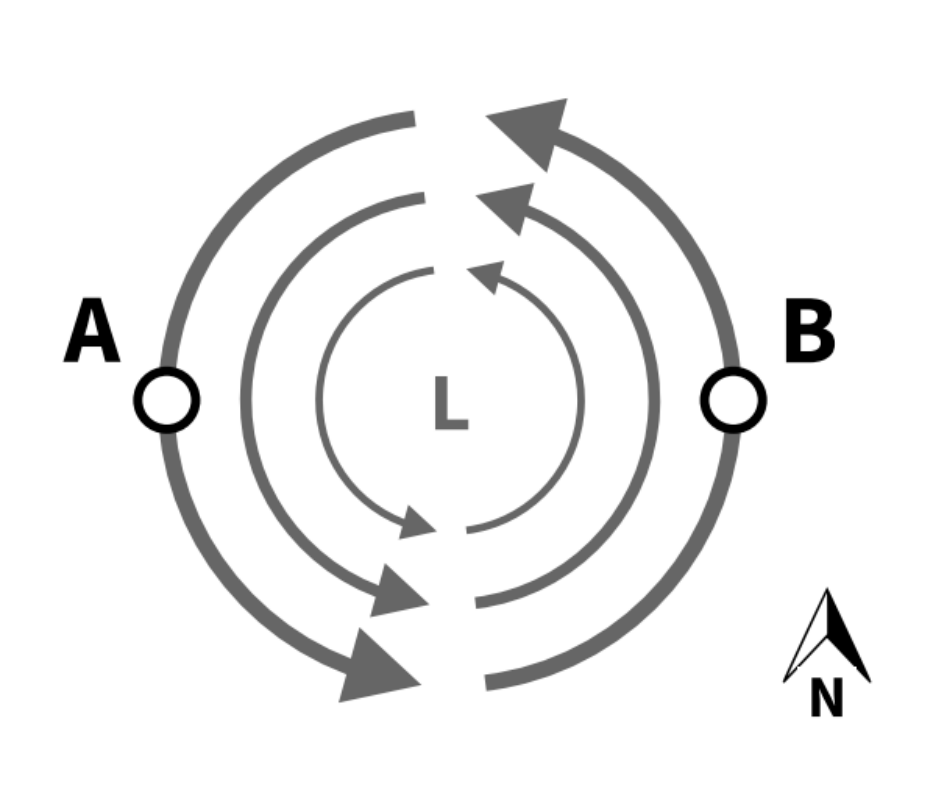
5.2.7 Storm motion

Winds such as easterlies advect the storm, but in addition, the **beta effect** works with it to move northwest in the northern hemisphere and southwest in the southern hemisphere. This is called the beta-advection model (2026 Camp exam). Note: It is essential to understand other advanced meteorology concept before understanding conceptually how this works, and this section is just for if you are curious!



Beta Gyres and beta advection model (Source: NOAA)

USESO Camp Exam 2026, Section I Question 15 Consider two points A and B on the westernmost and easternmost points, respectively, of a Northern Hemisphere cyclone. Each point on the cyclone has a relative vorticity ζ and planetary vorticity f such that $\zeta + f = \eta$, where the absolute vorticity η is conserved during the movement of an air parcel.



A. Relative vorticity increases at A and decreases at B, causing net movement to the north and west
B. Relative vorticity increases at A and decreases at B, causing net movement to the south and east
C. Relative vorticity decreases at A and increases at B, causing net movement to the north and east
D. Relative vorticity decreases at A and increases at B, causing net movement to the south and west

Solution: This is our beta-advection model. Since planetary vorticity increases with latitude, B must have decreasing relative vorticity while A must have increasing relative vorticity to compensate. This corresponds to answer choice A. As we already know, this causes motion to the north and to the west. **Answer: A**

5.3 Additional Info

5.3.1 Associated Hazards

Hurricanes don't only bring intense storms and intense winds, but they also bring other hazards:

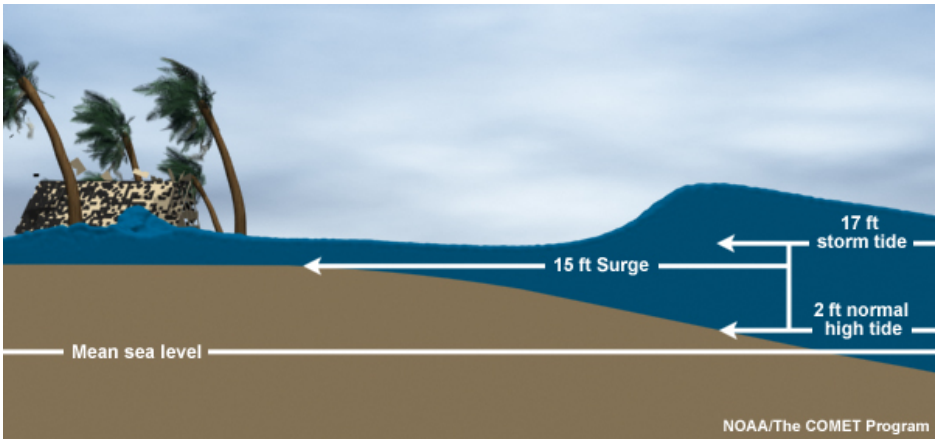
1. Tornadoes - Intense winds create wind shear as friction slows winds at the surface planetary boundary layer (creating a strong contrast), strong enough to develop mesocyclones, especially in the right front quadrant in the northern hemisphere.

2. Violent Waves and Rip Currents - The intense winds cause dangerous waves and the energy also leads to intense rip currents flowing back through sandbars.

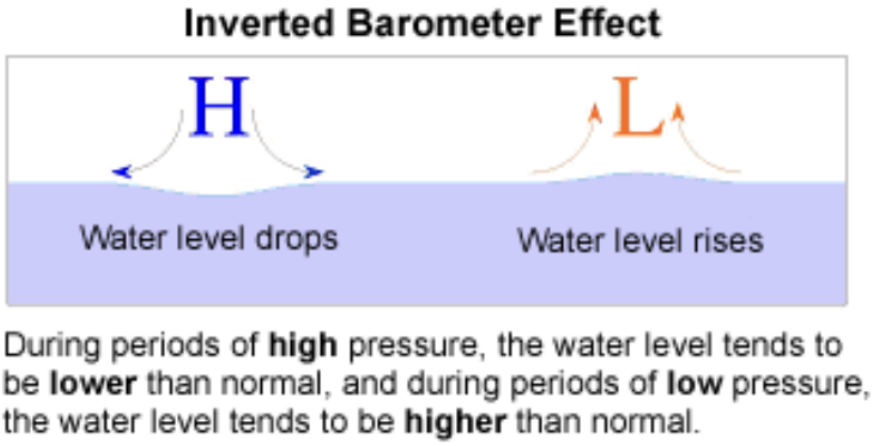
3. Flooding - This can happen due to a combination of storm surges, river discharge, intense precipitation, and other factors as well which is called compound flooding.

Let's focus in on one of the biggest hazards storm surges. Storm surges happen due to a combination of the **inverse barometer effect** where low pressure areas above water cause the sea level to rise (effect is 1cm per 1hPa) and then the more significant factor, wind stress (wind forcing the pileup of water). This can absolutely flood both regions directly on the coast and inland regions, and places like New Orleans with especially low elevation are very susceptible.

(Note: There are also things known as negative storm surges where the wind stress causes a lowering of sea level, and this usually happens if you are in one of the left quadrants of the storm).



Storm Surge coupled with astronomical high tides to form larger storm tides (Source: NOAA)



Inverse Barometer Effect (Source: Naval Postgraduate School)

5.3.2 Climate Effects

Rapid Intensification, where storms gain 30 knots in 24 hours, is becoming more common, especially in nearshore areas based on recent research. This makes regional risk assessment and predictability more difficult. Additionally, there's conflicting research and models on whether TC frequency will increase or decrease with climate change, but there's a consensus that **higher**

intensity storms will become more frequent. This also means more intense winds, storms surges, rainfall, and other associated hazards. Additionally, genesis locations will likely change due to climate change as water becomes warmer. Tropical cyclone will be able to be form from further up north in colder waters.

Additionally, climate oscillations like ENSO affect the hurricane season. El Nino decreases the likelihood of hurricanes in the Northern Atlantic and increases the likelihood in the Central Pacific and vice versa for La Nina due to wind shear.

5.3.3 Other

You should note that although tropical cyclones and extratropical cyclones are both cyclonic, they have very different mechanisms. Tropical cyclones are fueled by warm water (**warm core cyclones**) while extratropical cyclones are fueled by temperature contrasts between frontal boundaries (**cold core cyclones**). Warm core storms weaken with increasing height while cold core storms strengthen with height. Furthermore, tropical cyclones can even become an extratropical cyclone as it moves towards the mid latitudes.

Hurricanes are very complex, and there's much more to learn and a lot of research going on in topics like the inner core and improving different models, so I encourage you to explore more on your own.

6 Conclusion

Although there are a lot of types of weather and extreme weather, the fundamentals are the same with latent heat generating the energy required for the storms. However, it's also important to understand their differences as we discussed in this handout. Weather is the staple of meteorology, and it is important to understand weather to be better informed about our world.